
THIS IS ARMY MARS

January 2021

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DEPARTMENT OF THE ARMY
MILITARY AUXILIARY RADIO SYSTEM
FORT HUACHUCA ARIZONA 85613-7070

Message Through!

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IMPROVEMENTS

Chief Army MARS is the proponent of this publication. This manual is prepared at HQ, Army MARS.

Suggested corrections, or changes to this document, should be submitted through your State Director to the Regional Director. Any Changes will be made by the National documentation team.

DISTRIBUTION

Distribution is unlimited.

REFERENCES:

The following references apply to this manual:

1. DoDI 4650.02 – Military Auxiliary Radio System
2. AR 25-6 - Military Auxiliary Radio System (MARS) and Amateur Radio Program
3. ADRP 1 The Army Profession
4. ADP 6-0 Mission Command
5. AR 600-20 Army Command Policy
6. AR 530-1 Operations Security

SCOPE AND PURPOSE OF THIS MANUAL:

This manual is the authoritative and comprehensive document describing the organization of the Army MARS program, the Chief Army MARS' policies, requirements and expectations for MARS members. This manual condenses Army MARS policy into a concise digest and references applicable Army regulations. It assigns tasks, describes roles and imparts responsibilities to ARMY MARS volunteer leaders and Army MARS members.

This manual replaces and supersedes AM-1 This is Army MARS 31 MAR 2017, AM-3 Army MARS Operations Plan 4 April 2016, AM3 Annex B Operational Security Information Assurance and Privacy 6 July 2016, and portions of AM6 Annex C Technical Standards for MARS Station 16 June 2019. This manual is applicable throughout all of Army MARS and is superior to all guidance that may have or will be issued by any Army MARS Region or State.

INTRODUCTION TO ARMY MARS

Military Auxiliary Radio Service (MARS) is an Army sponsored program that organizes, trains, and utilizes Federal Communications Commission (FCC) licensed amateur radio operators, as military Auxiliarists, to contribute to and support the Department of Defense (DoD) (see fig 1). Army MARS is headquartered with U.S. Army Network Enterprise Technology Command (NETCOM) at Ft Huachuca AZ. NETCOM is subordinate to U.S. Army Cyber Command and provides the leadership, professional staff, organization and oversight to prepare Army MARS Auxiliarists to support DoD activities. NETCOM manages the Global High Frequency Enterprise Network (GHFEN) that Army MARS supports. The GHFEN provides controlled unclassified information using HF radio connectivity across the Army in the event of a denial of cyberspace.

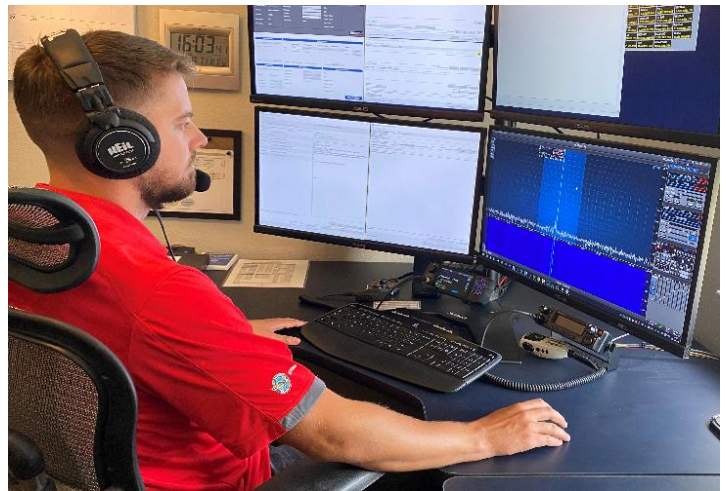


Figure 1 the 21st Century ARMY MARS member is a motivated and skilled individual who supports the United States

Army MARS is made up of private citizen volunteers who employ their own High Frequency (HF) radio stations in support of the DoD and Services. These individuals are organized into a military auxiliary, assigned internationally recognized U.S. Army call signs and operate on designated military radio channels. These Auxiliarists are organized by the Chief Army MARS, who is a Department of the Army Civilian, around Civilian Volunteer Leaders that assist the Chief with administration, training, and operational oversight of Army MARS.

MARS Auxiliarists are located in communities throughout the United States, its territories and possessions and serve as the principal interface between the DoD and licensed Amateur Radio operators in local communities. During a catastrophe and the immediate aftermath, local amateur radio stations are often the only source of communication, news, and information in the affected area.

The 21 Century Army MARS Member

Army MARS Auxiliarists are HF radio Subject Matter Experts. Their primary function is to provide contingency communications support to the DoD, the Services, combatant commands and their components.

Army MARS members are expected to mitigate risk in cyberspace, maintain operational and cyber security, recover quickly from a cyber-incident, utilize type three encryption, as well as establish reliable, military interoperable communications over long distances using HF radio when access to cyberspace has been impaired or denied. Potential Army MARS members must be capable of:

- Communicating effectively and concisely while being proactive, improvising and adapting existing guidance to new or unanticipated conditions.
- Utilizing effective HF radio stations and the effects of the ionosphere to communicate over long and short distances.
- Demonstrating skill in utilizing computer equipment in conjunction with radio transceivers; installing, securing, maintaining and updating of software, soft-modems, operating systems, configuring and securing local area networks without assistance.

Army MARS Auxiliarists demonstrate the Army's values of Loyalty, Duty, Respect, Selfless-Service, Honor, Integrity and Personal Courage. The world has changed dramatically in the nearly 100 years since the MARS structure was first developed and implemented. What has not changed is the dedication of the Auxiliarists who serve our nation.

The Army MARS Mission

Army MARS provides skilled Auxiliarists who are prepared to operate long distance HF radio networks during situations where the principal cyber network is degraded or unavailable, to support the mission of the Department of the Army and Department of Defense.

To clarify, Army MARS primary customer is the United States Department of Defense and the Services. The ARMY MARS mission, broadly stated, is to train and maintain civilian Auxiliarist contingency communication capabilities to appropriately support the US Armed Forces missions which include:

- Project power despite anti-access/area denial challenges.
- Operate effectively in cyberspace.
- Operate effectively in space.
- Defend the homeland and provide support to civil authorities.
- Provide a stabilizing presence.
- Conduct humanitarian, disaster relief, and other operations.



Figure 2 Army MARS member demonstrates the impedance of an antenna system

To be prepared to support this broad range of possible support, Army MARS Auxiliarists train on a variety of communication skills, whatever they may be, as is appropriate for a civilian auxiliary to perform. The communication skills and procedures needed to support the DOD are subject to change at any time. In providing this potential broad range of support, MARS Auxiliarists do not deploy. Auxiliarists provide the contingency HF network connectivity from their home stations. In some instances, select Auxiliarists may have the opportunity to conduct in-person, hands-on training with a military unit during small unit training. Auxiliarists will typically provide assistance with HF antenna configuration and siting considerations, radio procedures and distant end radio station support (see fig 2).

In those cases where there is normal access to space and or terrestrial telecommunications services, the requirement for auxiliary communications support of any kind is minimal. In most cases where a denial or impairment of space and/or terrestrial telecommunications is anticipated great efforts are made to mitigate the effects of the potential denial or impairment without utilizing civilian Auxiliarists. Conditions where a communications auxiliary would be utilized in an impaired or denied environment are continuously evolving.

The Army MARS mission consists of the following high-level tasks:

- 1) Provide radio communications support to the DoD by staffing the GHFEN with skilled HF radio operators and highly effective HF radio stations capable of long-distance communications utilizing multi-layer radio network techniques, including Radio Wire Integration to the Public Switched Telephone Network (PSTN) to assist units in the setup coordination of Joint Network Node (JNN) terminals and other requirements.
- 2) Support DoD Humanitarian Assistance and Disaster Relief (HADR) efforts by interfacing with Amateur Radio stations in disaster areas, or other Amateur Radio stations that are in contact with the disaster area, to obtain information concerning the conditions, impact and extent of the disaster, and other information pertinent to coordinating a possible U.S. HADR response.
- 3) Interface with members of the Amateur Radio Service in the communities throughout the United States, and to process situational awareness information products for DoD consumption.
- 4) Supporting unit training of cadets, Soldiers and guardsman in HF radio fundamentals by providing far-end radio contacts and serving as subject matter experts in HF radio techniques. MARS citizen-volunteers are uniquely qualified to contribute to improving the effectiveness of these military units' HF communication skills.

Amateur Radio supports most government and non-government organizations through direct individual initiatives at the local level. The DoD is the largest entity that Amateur Radio supports and has organized amateur radio nationally through MARS. While support to US military inside the DoD is often ad-hoc, Defense Support to Civil Authorities (DSCA) is always tasked in formal orders and limited in scope to a specific incident. Unlike most Amateur Radio emergency communication activities, MARS members do not seek out individual support

relationships with Civil Authorities, rather, act in a directed manner by military authorities at US Northern Command or their Service components when DSCA is authorized.

ARMY MARS Membership

Individual MARS Membership is open to civilian, volunteer Amateur Radio operators who desire to support the United States' defense related activities utilizing their Amateur Radio station.

Qualifications

Individuals applying for ARMY MARS membership must meet the following minimum requirements:

- Eighteen Years of age or older
- Citizen of the United States or lawfully admitted to the United States for permanent residence under 8 USC Chapter 12
- Hold an FCC Amateur Radio license (any class)
- Agree to follow all rules and procedures prescribed by the Chief Army MARS and have a desire to provide voluntary communications support to the defense of the United States
- Have no prior record of being terminated for cause by another Service MARS program
- Agree to the membership agreement, privacy statement, and non-disclosure statement in the Army MARS membership application
- Able to pass a background check, ie: Of good moral character, no felony convictions, or under indictment or information in any court for a felony, no associations with known terrorist organizations, not an unlawful user of illicit drugs, has not been discharged from the Armed Forces for other than honorable conditions, has not been terminated for cause by any MARS service, and has not been declared mentally incompetent.



Figure 3 Army MARS auxiliaries regularly exercise their skills and demonstrate readiness

FCC License

New Members must have a valid FCC issued amateur radio license and are then issued a provisional MARS Authorization To Operate (ATO). Completion of all initial training goals is required to renew the ATO. The radio station call sign issued to an Amateur Radio operator by the FCC is not valid on MARS frequencies nor can the MARS call sign be used on Amateur Radio frequencies.

Initial Training

Once accepted into ARMY MARS, new members are assigned sponsors and must complete an orientation training course in their first six months of membership.

Hands-On training is on a telephone conference bridge or computer collaboration platform and tasks are performed on-the-air using the trainee's radio equipment. Trainees must complete a task list demonstrating proficiency of various radio operator skills.

Activity Reporting

ARMY MARS members are expected to meet a minimum, on-going training requirement to retain their skills. To assess readiness, monthly reports are required from all members. These reports indicate the status of the members' ARMY MARS station and time spent training for MARS activity.

A minimum of 72 hours per year (optimally spread over 18 hours per quarter) and active participation in at least one national level exercise is required of MARS members to retain membership. Amateur Radio activity does not meet this requirement.

Members who do not provide reports for three consecutive months are considered to have abandoned their membership and will have their MARS authorization to operate cancelled.

Members are granted up to 90 days leave of absence from MARS on a case-by-case basis through their State MARS Director. Leaves of absence or extensions over 90 days may be approved by the Region Director.

Other Requirements for MARS Members

The Department of the Army requires that its MARS members have access to equipment suitable to accomplish the tasks the Army requires of its MARS stations. The member need not own the equipment, or have it in their house, but must have unrestricted and ready access to the equipment at any time so they can train for and contribute to the mission of the Department of the Army on a regular and ongoing basis. The following are required:

- A radio station capable of transmitting at least 100 watts and capable of being modified to operate between 2-30 MHz on both SSB voice and MIL-STD 188-110A serial PSK mode. The transmitting equipment must meet minimum technical parameters required by the National Telecommunication Information Agency (NTIA). (ref NTIA Manual section 7.3.9)
- A computer system connected to the radio equipment capable of operating the supplied messaging software, for drafting, sending and receiving required messages.
- An internet connection with email and web is required for administration and a telephone is required to access telephone bridges utilized for training.

On an on-going basis, ARMY MARS members will be required to perform the following tasks:

- Check into a MARS net. Most members check into their State or Region MARS net at least once per week. Members are also required to perform the Net Control Station function; and transmit and receive formatted message either by voice or data mode.
- Maintain a radio station log of all on-air MARS activities.
- Maintain and secure computer systems and local area networks, install and update software without assistance.
- Using the supplied messaging software, receive and send messages using MIL-STD waveforms and message text formats. Encode and Decode messages.
- Train in at least one national level DoD exercise annually. There are at least four national level exercises each year.
- Meet all performance standards established by Chief Army MARS.

Membership Term

The ATO is good for one calendar year and must be renewed annually. New members in training receive a six-month authorization that may, depending on the date it was issued, extend into the next calendar year. Members are expected to keep current their membership record with the Army MARS Headquarters and report of any change in address, telephone number or email.

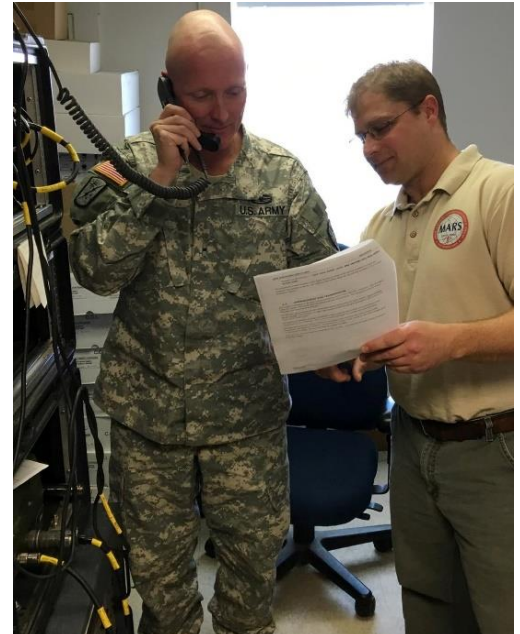


Figure 4 MARS Auxiliaries are expected to be HF radio Subject Matter Experts and are uniquely qualified to assist military personnel when exercising skills.

THE ORGANIZATION OF ARMY MARS

The US Army Cyber Command (ARCYBER) is the Department of the Army (DA) executive agency for ARMY MARS which is tasked with directing, managing and funding the ARMY MARS program. US Army NETCOM is the component of ARCYBER to which ARMY MARS is assigned. The Commander ARCYBER appoints a Department of the Army civilian as the Chief Army MARS to execute the HQDA responsibilities of the Army MARS program (see fig 5). The specific responsibilities of ARCYBER and the Chief are described in Army Regulation (AR) 25-6.

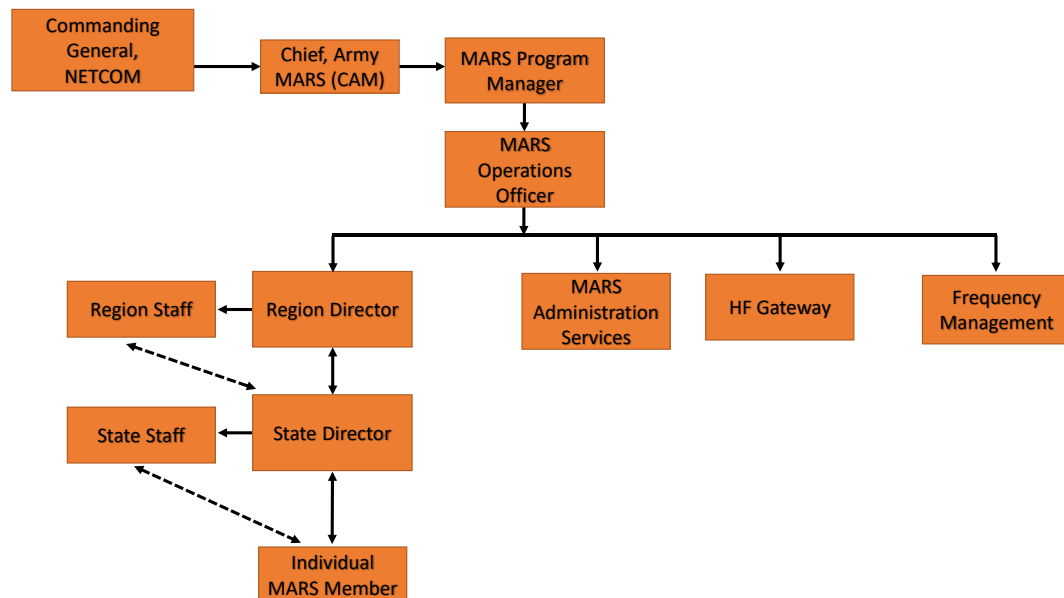


Figure 5 Army MARS Organizational Structure

The Chief Army MARS has organized Army MARS into the following tiered system:

National Leaders and Staff

- Chief, Army MARS.
- In addition to the Chief, other government leadership and contract staff are assigned in a support role to the MARS program.
- Auxiliarists support the Chief by serving as Region Directors, State Directors, national net managers, liaison officers, and assisting with other staff functions such as administration and training. The Chief may, at his/her discretion, assign volunteers to various positions at the national level to perform such tasks as required.
- The MARS Governance Executive Group (MGEG). This is a work group comprised of Region Directors and other select advisors which advises the Chief Army MARS on policy and other matters.

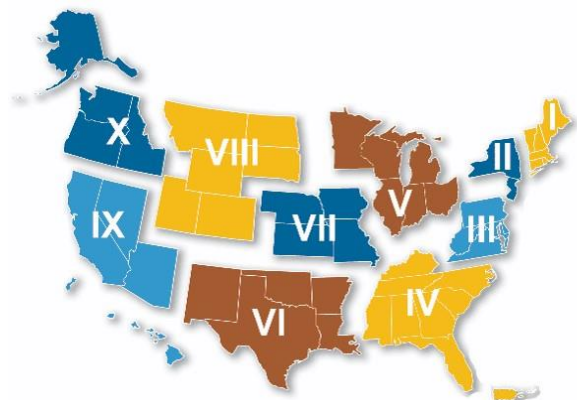


Figure 6 Army MARS Regions

Region Civilian Volunteer Leaders

The Chief appoints volunteer leaders to serve as Region Directors of the 10 MARS administrative regions and the National Support Network. These regions coincide with the 10 Federal Emergency Management Agency (FEMA) regions except that Puerto Rico and the Virgin Islands are assigned into MARS Region 4 (see fig 6).

The Army MARS Region Director implements the policy and directives developed by Chief Army MARS, oversee the Administration of the Auxiliarists, exercise tactical control of MARS operations in the region, and assess the readiness of the MARS program in their region. In addition, the Region Director is primarily responsible for establishing high level training objectives, assessing training and operational readiness, and the execution of the collective tasks in support of the national mission.

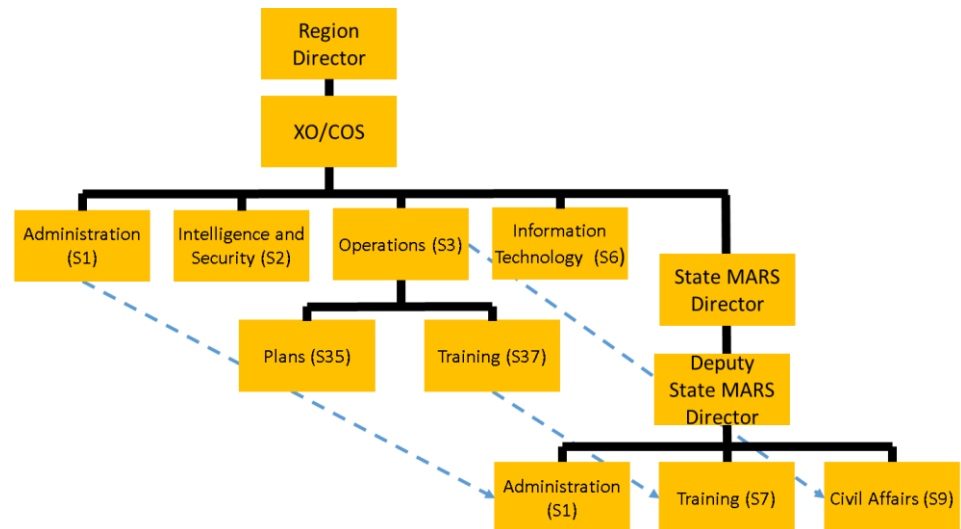


Figure 7 Organizational Structure of the Region and State

The Region Director, with the assistance from region staff and State Directors, will assess the capabilities within the region to ensure state Army MARS organizations (or other subdivisions of the regions that may be established) are trained and organized to effectively carry out these collective tasks (see fig 7). To achieve these objectives, the Region Director will:

- a. Establish a leadership team of Army MARS members from the Region, delegate authority to the members of the leadership team as required and establish goals, objectives and milestones to monitor the team's progress.
- Appoint an Executive Officer or Chief of Staff, an Operations Officer (S3), an Administrative Officer (S1), and an Information Technology Officer (S6).
- Use the Executive Officer/Chief of Staff to track all ongoing initiatives and routine staff activity in the Region, the respective goals and objectives, and track progress, milestones, and ensure all actions are progressing according to the Region Director's expectations.
- Appoint a State MARS director to each State where there are sufficient Army MARS members, or a combination of States where there are sufficient Army MARS members, to administer and train to standard the individual Army MARS members in the State.
- The Operations Officer may appoint a Training Officer under the Operations Section to access the membership, oversee training activities in the Region, and ensure the Region's Mission Essential Tasks are trained to standard. The Operations Officer will supervise the Training Officer, who has direct liaison authority to the Training Officers assigned in each State.
- The Operations Officer may appoint a Plans Officer under the Operations Section to draft Region operations order and fragmentary orders, the Region Operations Plan, and other appropriate tasks as required.
- Instruct the Operations Officer to exercise Direct Liaison Authority/ Direct Line of Authority (DIRLAUTH) with the State MARS Directors to ensure Civil Affairs coordination and productive relationships with local Amateur Radio Clubs and Service Groups are in place, and to ensure all relevant and current Civil Affairs related information is forwarded to the Plans Officer.

- Instruct the Information Technology Officer (S6) to organize appropriate members of the Region to establish a Military Standard Communications (MSC) software help desk and ensure relevant Information Technology capabilities are fully functional and up to date with the current baseline.
 - Charge the Administrative Officer (S1) with overseeing the annual membership renewal, ensure all member's address and contact information is current, and forward the relevant data to HQ Army MARS as required.
 - Instruct the Administrative Officer (S1) exercise DIRLAUTH with the State MARS Directors to ensure the members are in compliance with all administrative requirements, new member actions are completed in a timely manner, collect each State's reports and file the Region's monthly readiness report.
 - Instruct the Operations Officer to schedule Region MARS HF radio operations on a daily basis in minimum Sustained Network Operations Posture, and direct the appropriate members to serve a net control stations as required.
- b. Direct Auxiliarists to support tasked missions and assignments.
- c. When opportunities exist, direct qualified personnel to assist in training counterparts in the Regular Army, Reserve and National Guard on HF radio procedures and techniques. Coordinate with the military component to schedule support and training during their duty hours. The goal is to demonstrate radio techniques and procedures using the unit's assigned equipment and not demonstrate/showcase the MARS Auxiliarists' equipment.
- d. Assign Auxiliarists in the Region to support to the Trans-Global net to maintain regular HF radio (both single channel and automatic link establishment) contact throughout the MARS enterprise.
- e. Develop and improve working relationships with adjacent activities (eg. AFMARS, Amateur Radio Service, ARRL, and Radio Amateur Civil Emergency Service where applicable). Establish Region Joint MARS Operations Groups (RJMOG) with Air Force MARS when directed.
- f. Coordinate and facilitate MARS Auxiliarists' registration in and support to the Department of Homeland Security (DHS) SHARED RESOURCES network (SHARES).
- g. Identify personnel with the capability/equipment to support the National Support Network (NSN).
- h. Review disciplinary procedures, suspensions, and terminations for cause. Ensure disciplinary procedures are documented as required by Chief Army MARS.

State Civilian Volunteer Leaders

The State MARS Directors are the first line supervisors who work directly with the individual MARS volunteers to lead, organize, train, and counsel members. The State Director appoints a staff consisting of An Executive Officer (XO), and Administrative Officer (S1), a Training Officer (S7) and a Civil Affairs Officer (S9), to assist with the administration, and individual training of the ARMY MARS members in the State, ensuring each auxilist is "mission ready".

The principal responsibilities of the State MARS director and the staff are:

- Vetting, orientation, counseling, and training of new volunteers.
- Assisting HQ Army MARS with reconciling the national member database and supporting the Region Administrative Officer with the annual membership renewal.
- Maintaining discipline, counseling members, and initiation of disciplinary procedures in accordance with AR 25-6.
- Approving requests for leave of absences from Army MARS volunteers.
- Identifying Army MARS volunteers who are eligible for awards and submitting members for award recognition.
- Initiating processes for transfer of MARS membership.
- Processing forms for a new MARS ATO or renewal.

The state director assists the Region Director in planning and coordinating civil support activities in the state and may act as liaison to the Region Director for the National Guard in the State.

The state director, assisted by the Civil Affairs Officer, ensures close coordination and productive relationships are established with Amateur Radio groups in the state that are capable of providing assistance to accomplish the Army MARS mission during exercises and contingencies. The state director provides the region director with principal points of contact for local amateur radio leaders and amateur network schedules for inclusion in the Civil Affairs section of the region operations order as may be required.

The state director, assisted by the staff, ensures the region director has operators who are proficient on the collective tasks assigned to the Region. Additionally, the state director, assisted by the training officer, will continuously evaluate the proficiency of all individual tasks through regular training and evaluation, to ensure the standards are met by the MARS volunteers within their organization.

To achieve this, the state director, assisted by the staff, will ensure:

- a. The continuous evaluation of the readiness of the State Army MARS organization with routine readiness reports available to the Region Director as required.
- b. The continuous evaluation of the effectiveness of the membership to perform the individual tasks with routine readiness reports available to the Region Director as required.
- c. The delivery of routine training to MARS volunteers in order to achieve a high level of competency in the required individual tasks.
- d. Distribution to the Army MARS volunteers in the state the required documentation and software to execute the individual tasks.
- e. The timely distribution of information, messages, and directives from the Chief Army MARS and HQ Army MARS to the membership.
- f. Encourage participation in SHARES (CONUS, Alaska, and Hawaii).
- g. Establishment of a telephone conference bridge or computer collaboration capability for collaboration and training.

Army MARS Volunteers

Maintains all required capabilities and participates in ongoing training and exercises as required.

Utilizing MIL-STD waveforms and message texts; High Frequency radio equipment on authorized frequencies; and in accordance with AR 25-6, execute taskings as received from Chief Army MARS as communicated through State and Region Army MARS Directors.

The Army MARS Manual System

- The AM 1 manual, this manual, describes the Chief Army MARS policy for the organization and operation of the Army MARS program.
- AM 3 series manuals describe the operations plans of the specific missions assigned to ARMY MARS
- AM 6 series manuals describe the operating instructions and policies specifically related to the radio network.
- AM 7 series manuals describe the basic and ongoing training program.

CONDUCT OF ARMY MARS MEMBERS

The Army MARS Ethic

As an all-volunteer Army auxiliary, civilian volunteer MARS members are held to the same core values and principles as our Soldiers and Department of Army civilians: loyalty, duty, and respect.

All Army MARS Auxiliarists are leaders and civilian ambassadors for the Army and therefore are expected to all times:

- Carry themselves as examples of virtue, honor, integrity, and honesty
- Be vigilant in all that you do as each of you are reflections of the Army...even as civilian volunteers
- Take necessary action to watch for and help eliminate illegal, immoral and unethical practices both on the air and in person with your fellow MARS members.
- Report possible violations to your volunteer chain of command or use the Chief Army MARS "open door" policy to report your concerns if they cannot be resolved at a lower level.

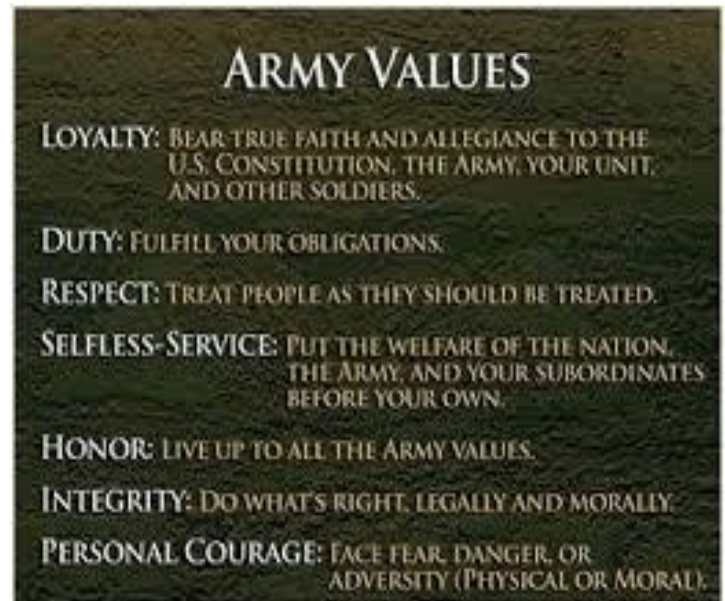


Figure 8 Army Values

The desired end state is a climate where all members able to thrive in a healthy environment in order to carry out the MARS mission and execute the Chief Army MARS' intent.

Prohibited Behaviors: Hazing, Bullying, Harassment, and Discrimination

The Army and Army MARS is a values-based organization where everyone is expected to do what is right by treating all persons with dignity and respect. Hazing, bullying and other behaviors that undermine dignity and respect are fundamental in opposition our values and are prohibited.

Hazing: Any conduct whereby a member(s) recklessly or intentionally causes another member to suffer or be exposed to any activity that is cruel, abusive, humiliating, oppressive, demeaning and harmful. Soliciting or coercing another to participate in any such activity is also considered hazing. Hazing can be verbal or psychological in nature and may be committed through written or phone messages, text messages, email, social media or any other virtual or electronic medium.

Bullying: Bullying is any conduct whereby a member(s) intends to exclude or reject another member through cruel, abusive, humiliating, oppressive, demeaning or harmful behavior, which results in diminishing the other member's dignity, position, or status. Bullying may include an abuse of authority. Bullying tactics include, but are not limited to, making threats, spreading rumors, social isolation, and attacking someone physically, verbally, or through use of electronic media.

Sexual harassment is a form of gender discrimination that involves unwelcome sexual advances, sexual innuendo, request for sexual favors, and other verbal or physical conduct of a sexual nature. Sexual assault and sexual harassment are not the same, although they are related to each other.

The Motivation to Serve

Army MARS is a unique activity for most people, requiring a great deal of personal investment for very intangible, perhaps even esoteric personal reward (see fig 9). MARS members do not receive surplus government equipment nor are they provided with radios or reimbursement. Most MARS members find their service rewarding for three primary reasons:

- the knowledge they are performing a patriotic duty doing something positive that is of essential value to their nation's defense,
- the camaraderie of serving with other MARS members and the support that comes with that association,
- honing and refining contingency communications skills in support of the Department of Defense and the Services.

The first standard of conduct for all Army MARS auxiliaries is to safeguard these three primary rewards which are essentially the fuel that energizes Army MARS members. MARS Leaders at all levels must establish a positive climate that facilitates team building, encourages initiative, and fosters collaboration, dialog, and mutual trust and understanding. Leaders balance uncertainty, remain flexible, and provide a positive climate where MARS Auxiliaries have the opportunity to excel.

It is incumbent on all MARS auxiliaries to safeguard the MARS experience for our fellow auxiliaries by channeling internal disagreements into appropriate, private communications with leaders and not permit inappropriate discussions to occur in conjunction with MARS operations and training. Most members did not join MARS to participate in negativity, conspiracy theories or politics; those types of occurrences will diminish everyone's MARS experience. It is all our duty, as leaders and members, to support our colleagues by establishing a positive, productive climate with our teammates.



Figure 9 Service is often its own reward

Operational Security

Many of the Tactics, Techniques and Procedures (TTP), including technical details of radio circuits and technology utilized in MARS is dictated in accordance with DoD requirements. The details concerning many of these requirements are not provided to MARS members to prevent revealing information regarding the readiness and operational capabilities of US military forces. While MARS members are never exposed to classified information, they might learn enough unclassified facts to speculate and arrive at their own conclusions (see figure 10). Some persons observe MARS for the purpose of aggregating unclassified facts seek to gain an unauthorized insight into US capabilities. This section on Operational Security (OPSEC) is intended to guide members to minimize exposure of Controlled Unclassified Information (CUI).

OPSEC provides a methodology to manage risk. It is impossible to avoid all risk and protect everything, and some risk must be accepted to do our job in MARS. Therefore, OPSEC is a balance. Practicing good OPSEC is a state of mind, a process of continuous evaluation and decision-making that balances risk, the need to know, and the overriding imperatives associated with achieving operational or strategic objectives.

These fundamental assumptions about OPSEC apply to ARMY MARS:

- MARS members should never be exposed to or hold Classified information. Likewise, members should never speculate on potentially classified information based on aggregating unclassified facts.
- Good OPSEC should not prevent MARS members from doing their job. MARS members are encouraged to accept risk where it is necessary, and avoid it where there is little to gain. For instance, in some cases MARS members are required to identify themselves as MARS members to local officials or amateur radio operators in order to conduct operations, exercise, or recruit members. This is acceptable risk.

- Be aware of your environment, how what you say may be overheard by third parties and re-distributed by them. Avoid communicating potentially sensitive information in writing outside the organization, including signature lines, websites, and forums. Personal communications, typically one on one, are less likely to be observed by third parties and reproduced.
- OPSEC is not a set of rules, an encryption system, a document marking or classification. OPSEC is a State of Mind resulting in good, balanced decisions based on the totality of the circumstances and unique requirements of the mission.
- Some things simply cannot be protected in the MARS environment. For instance, your voice may be recognized while operating on a MARS frequency by your Amateur Radio friends and because of that they discover your MARS call sign. Should a specific threat or methodology to exploit MARS become known, additional guidance will be provided.
- OPSEC does not justify "I've got a secret" behavior. Act with humility.
- Good OPSEC depends on human behavior



Figure 10 MARS Auxiliaries may at times work alongside Active Duty, Reserve and National Guard. Discretion about sharing sensitive information observed during your MARS experience is the foundation of OPSEC.

Information Assurance

MARS members can potentially hold sensitive information, or send or receive it using private email. There are strenuous rules governing Information Assurance (IA) in the US military on government owned computers and networks. The following special considerations apply to MARS.

- A MARS member's environment is usually a private residence, a personally owned computer, and commercial (dirty) internet. It is not possible to secure this environment in the same manner as a government owned facility. A lower expectation of Information Assurance and physical security must be accepted for this environment.
- Privately owned personal computers connected to the internet can be monitored, disabled, or controlled by unauthorized third parties even when modern security products are properly employed.
- Encryption is of little value when the computer systems in which sensitive information is being processed may potentially be monitored by, under the control of, or may be disabled by unauthorized third parties. Therefore, it is critical that computers that process sensitive information are disconnected from the internet at all times.
- Encryption is only as secure as the personnel who hold the key.

AMARS members are encouraged to employ a second computer system that is normally not connected to the internet and contains only the required software to operate in MARS. Since this computer is not internet connected, it does not require frequent operating system updates, patches or security products. This is referred to as a "stand alone computer", and often it may be an older model and still run MARS software successfully.

Controlled Unclassified Information

Controlled Unclassified Information (CUI), formerly known as essential elements of friendly information, is defined as information important to the successful achievement of U.S. objectives and missions, or which may be of use to an adversary of the United States. CUI consists of specific facts about friendly capabilities, activities, limitations (includes vulnerabilities), and intentions needed by adversaries for them to plan and act effectively so as to degrade friendly mission accomplishment. The document marking "CUI" replaces For Official Use Only (FOUO) in all Army MARS documentation.

Information that HQs Army MARS determines to be CUI may not be shared outside the organization, publicized on websites or discussed in public places. This information may or may not contain document markings that identifies it as CUI, and all members must be able to recognize this information as CUI without assistance.

The following information is determined to be CUI by the Chief Army MARS, and is to be protected according to OPSEC policy by all ARMY MARS. The Chief may issue a memorandum each year updating this list:

- Any material portion-marked with FOUO or CUI such as MARS frequency listings
- MARS call signs shall not be listed with any other amateur radio or General Mobile Radio System (GMRS) call signs on badges, signature blocks, websites or any other written forum, medium or conveyance.
- MARS member rosters that list name, address, phone number, and email address (this information is protected even though it does not meet the definition of PII).
- All digital messages transmitted via HF radio shall be online encrypted regardless of message content. Exceptions to this are simple test messages used to troubleshoot software or messages transmitted to active duty, reserves, and National Guard units when they do not have a compatible crypto capability.
- Specific locations, unit names or any information concerning units or headquarters that MARS is supporting. Examples: Acceptable, U.S. Northern Command (NORTHCOM), U.S. Strategic Command (STRATCOM), Pennsylvania Army National Guard; Unacceptable, 3-124 Infantry Division located in training area xx at Fort Irwin, California in preparation for deployment to the middle east in two months.
- Contingency HF communications being provided to a specific unit at a specific location highlighting that unit's communications equipment is non-mission capable.
- Specific dates for any DOD COMEX or NORTHCOM DSCA exercises to include pre-exercise coordination telcon dates
- Authentication tables, if issued

Protecting Personal Information (PII)

PII is information that links, relates, identifies, or unique describes a specific individual. Most individuals make decisions every day about sharing their own personal information with others. It is an individual's right to manage their own personal information in the manner they choose; however, it infringes on an individual's right to share their personal information without their authorization. Securing PII is centered on protecting other people's private information.

All Army MARS members, in the membership agreement, agree to have the personal information from the application form available to other MARS members, the US military, and the US Federal Government.

PII is information that if lost or compromised could potentially result in harm or identity theft. Some examples of PII are listed below.

An individual's Name used in conjunction with:

- Social Security number, full and truncated
- Driver's license and other identification numbers
- Citizenship, legal status, gender, race/ethnicity
- Birth date, place of birth, age
- Religious preference
- Security clearance
- Mother's middle and maiden names
- Spouse information, marital status, child information, emergency contact information
- Biometrics
- Financial information, medical information, disability information
- Law enforcement information, employment information, educational information
- Military records

MARS Member Records

Army MARS HQs solely maintains records on Army MARS members. The membership database is specifically designed to avoid collecting and storing a member's PII. Region Directors, State Directors and their staff forward information such as application forms, copies of amateur radio licenses, and training certifications, electronically to HQ, Army MARS. Generally, this information does not contain PII.

Army MARS leaders may keep a roster of current MARS members in their area of responsibility. For the purposes of accessing and reporting the status of readiness, training and member participation, Army MARS leaders make notes of their observations. Once no longer needed, notes, email, and other documentation that would, if retained, comprise an unauthorized and illegal personnel file must be destroyed.

Use of Army MARS Call Signs, Titles, and Signature Lines

MARS call signs are used to establish and maintain communications between radio stations. Your Army MARS call sign belongs to the Department of the Army and is assigned to the radio station you operate for on air

identification. It is not a personal call sign like your Amateur Radio call sign that is granted to you by the FCC. While your membership in MARS is not secret, the guidance on use of radio call signs is intended to provide some measure of anonymity between your on-air persona and your actual identity.

Use MARS radio call signs only when communicating by radio.

Army MARS call signs should not be used in signature lines, email addresses, name badges, etc.

To most people, a radio call sign appears to be a meaningless jumble of letters and numbers. This is useful while operating on the radio but distracting and unprofessional in personal interactions, even among other MARS members. Call signs are out of place in telephone conversations or personal meetings.

Your full name and Army MARS job title is how you should identify yourself in MARS related correspondence and MARS related in-person interactions when your name alone does not suffice. This is a plain language statement of your affiliation to Army MARS and your position. Signature lines are not required for ARMY MARS members and should only be used in MARS related correspondence. In those cases where members choose to use them for MARS related correspondence, the following simple standard applies:

The signature line for an Army MARS member:

Mr. Alfred E. Neuman
U.S. Army Military Auxiliary Radio System Member

The signature line for an Army MARS officer:

Ms. Diana Prince
U.S. Army Military Auxiliary Radio System
Region One Training Officer

This practice ensures all parties have a clear understanding of who you are and your roles and responsibilities.

HF RADIO STATION BEST PRACTICES

Location of HF Radio Stations

Not all locations are suitable for HF communications. Many residential, home owner association, apartments or commercially zoned areas are not suitable due to local regulations and civil contracts that prohibit certain activities such as the construction of antenna systems. At some locations there are high levels of electromagnetic interference (EMI) which cannot be mitigated. Some residences lack sufficient outdoor space to construct an effective antenna system. MARS stations located in one of these areas will likely have difficulty completing the mandatory training and participation requirements of the MARS program, and will not be able to provide useful support to the mission of MARS. MARS members considering a station in one of these locations should seek an alternate location to operate, such as an amateur radio club station.

RF Exposure Limits, Regulation

Antenna systems must be located to ensure personnel and life sustaining equipment are not exposed to harmful radio frequency radiation or subject to harmful interference.

Maximum Radio Frequency exposure limits for personnel are required by law and are set in 47 CFR 1.1310.

Near Vertical Incidence Skywave vs Long Range HF Communications

Near Vertical Incidence Skywave (NVIS) communications is typically defined as HF radio communications, using a vertical or nearly vertical takeoff angle, that provides omni-directional coverage to a distance up to 150 miles from the transmitting station.

A horizontal dipole antenna less than $\frac{3}{10}$ a wavelength above ground and less than $1\frac{1}{4}$ wavelength in length, will maintain an omni-directional radiation pattern, within 3db variation, between 90 degree to 60 degree "near-vertical" take off angles. A 60 degree take off angle will provide communications at approximately 150 miles distance, using the F2 layer. For communications more than 150 miles, a lower take off angle is required, the directional attributes of the horizontal dipole apply, and a different antenna configuration that favors lower take off angles is recommended.

The primary function of the AMARS HF radio network is to provide long distance radio communications, up to 1000 miles or more. Typical operating distances are 300 to 600 miles for operation in one of the larger Regions, and significantly more for relaying messages into and out of the Region.

Grounding, Bonding, Shielding

The National Electric Code sets standards for residential grounding of AC power systems. Motorola standard R56 sets standards for grounding Land Mobile Radio communications systems. UFC 3-575-01 and MIL HBK 419 provide general grounding and specific instructions for electronics equipment. UFC and MIL HBK 419 are available at www.usarmymars.org.

Grounding, bonding and shielding are critical for safety when operating electronics equipment, and essential for proper operating and protection of the equipment. Electro Magnetic Interference (EMI), Radio Frequency Interference (RFI) and intermittent or poor performance of transmission and antenna systems are all effected by good grounding, bonding and shielding. A review of the grounding, bonding and shielding systems should be one of the first steps in troubleshooting radio/telecommunications equipment.

All MARS members are expected to install and/or maintain electrical, electronic, telecommunications and associated equipment according to relevant standards. Failure to do so may result in poor performance and unsafe conditions.

Station Power and Back-Up Power Considerations

While not a mandatory membership requirement, backup power is essential for communicating during emergencies where electrical service is interrupted potentially for long periods of time. MARS members are encouraged to incrementally improve their backup power capabilities over time.

Backup power systems utilized by MARS members include:

Generator:

Advantage: Generator Systems provide relatively high current capacity and are suitable for larger loads. Generators, with a proper transfer switch, can use much of the existing AC wiring and infrastructure.

Disadvantage: Generator systems are often heavy, noisy and use large quantities of fuel (usually propane, natural gas, diesel or gasoline). The fuel products may be unavailable or in short supply during a national emergency so run time is potentially limited to fuel on hand. For incidents that span a very long period of time, it is likely the stored fuel supply will be expended. Untreated fuel has a limited storage life. Internal combustion engines require regular maintenance for reliability.

Safety: Generators use flammable fuels that pose health dangers to people exposed to vapors or exhaust.

Solar:

Advantage: provides DC current for charging batteries or operating small electronic systems. Larger systems with inverters, battery banks, and transfer switches can supplement AC mains power. While there is sunlight, Solar is self-sufficient and will continue operating throughout the lifespan of the panels. Solar panels themselves are usually easy to clean and maintain, and are reliable.

Disadvantage: Limited in capacity. Larger systems become very expensive. Solar power is only generated in good sunlight – and must be supplemented with storage or other methods of generation.

Safety: High DC voltages or currents may be present when many panels are wired together.

Wind:

Advantage: Provides DC current for charging batteries or operating small electronic systems. Is a good supplement for solar as it will often generate power even when there is little sunlight.

Disadvantages: Mechanical parts require regular maintenance. Only generates when wind is blowing. Potential for noise and RFI.

Safety: Turning blades are a hazard to personnel in close proximity.

Vehicle Electrical Systems:

Advantage: Vehicle battery and backup systems are probably the easiest and most common to come by. Almost everyone has one sitting in their driveway/parking lot. They will provide DC power to small electronic loads.

Disadvantage: Requires motor fuel to continuously operate. The vehicle engine must be started before the battery becomes discharged, and run long enough to become fully charged. Vehicles may also be required for other tasks. Fuel for vehicles may be unavailable, and what fuel is on hand may be reserved for high priority transportation.

Safety: Under hood has several hazards, including spinning fan blades, very hot surfaces, exposure to chemicals and vapors. Care should be taken to ensure the vehicle remains in park, wheels are blocked, and the vehicle cannot move when the engine is running.

Battery Backup:

Advantage: Battery backup is a relatively easy way to start a DC power backup “system”. A pair of deep cycle marine batteries, a DC power supply set to the float voltage, and a pair of high current diodes is a simple and effective system that will support up to moderate DC loads and re-charge during even brief periods when mains power is restored. DC Battery backup can be supplemented with wind, solar or generator systems, and over time evolve into complex and capable backup power systems.

Disadvantage: Must be re-charged to continue operating. In situations where mains power is not cycled on for a few hours each day, some other means of charging the system is required. Lends itself well to an evolution into a more complex system, but not as a standalone. Some battery types must be maintained and cleaned regularly. Disposing of batteries may be subject to hazardous materials (HAZMAT) regulations.

Safety: Batteries by their nature are hazardous. Lead Acid and AGM batteries are heavy, and contain acid. Maintaining batteries includes exposure to caustic and potentially hazardous chemicals. Hazardous fumes or vapors may be present when charging or discharging. Risk of fire and explosion is present. Other battery types present similar hazards. Note explosive hazards and hazmat for lithium batteries.

ANTENNA SYSTEMS, BEST PRACTICES

To be safe and effective, antenna systems must have a good grounding system, be constructed high above the ground, and in the clear away from people, obstructions, structures, buildings and power lines.

HF antennas should be placed away from occupied structures (such as residences) and other sources of EMI. EMI will degrade the signal to noise of received signals and require the transmitting station to use much higher power than would otherwise be required to achieve communications. The responsibility for overcoming EMI is with the receiving station and is achieved by eliminating the source of the noise.

Antenna system should be capable of transmitting and receiving effectively on all frequencies throughout the HF spectrum.

Although MARS stations usually train with other MARS stations located nearby, the desired capabilities for MARS stations requires effective HF communications for distances greater than 500 miles. Antenna systems that exhibit low angle of radiation are preferred over antennas which emit their energy at a near vertical angle of incidence.

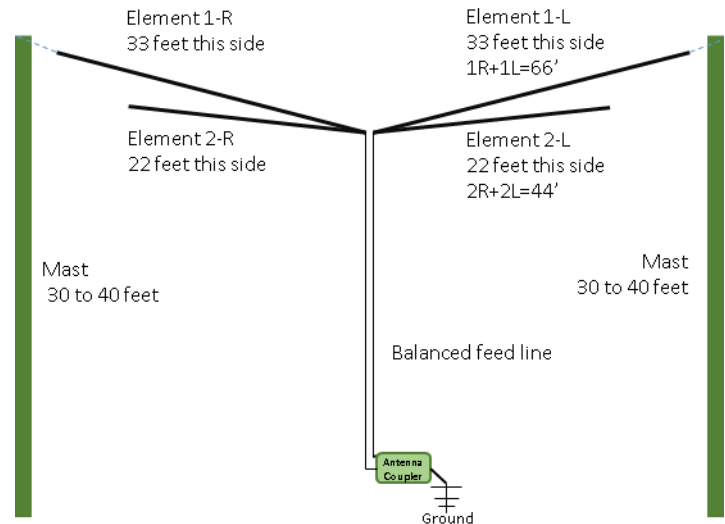


Figure 11 Two Wire Non-Resonant Dipole Antenna for use on multiple frequency bands

Non-Resonant Dipole Antenna

Most Amateur Radio operators are familiar with the dipole as a resonant antenna.

The dipole is a general-purpose HF antenna suitable for both long distances and short-range communications. The dipole antenna exhibits multiple radiation patterns, depending on the operating frequency and length in wavelength. When the antenna length is 1.5 wavelength or less relative the operating frequency, the antenna will exhibit a bi-directional pattern. When the antenna is more than 1.5 wavelength relative the operating frequency the antenna will exhibit four or six lobes in a more omni-directional pattern. The dipole antenna should be used when communications with stations in several different locations is anticipated, and the situation allows two relatively high supports, exposed in plain view, and clear of other structures. Dipoles work best when they have clear view of the sky and horizons, high above the ground and are located clear of nearby structures.

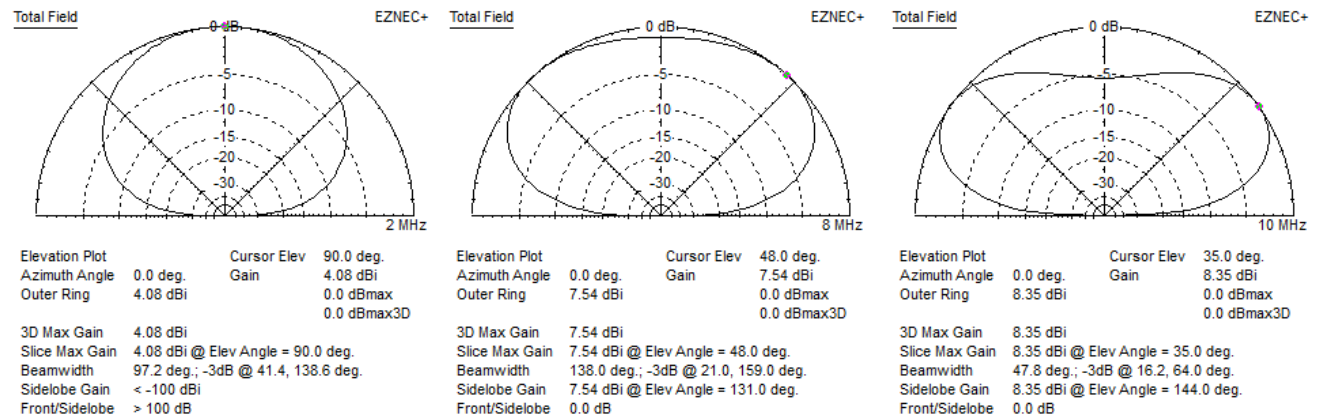
At a height of 40 feet above the ground, the antenna will exhibit a good high angle NVIS pattern on radio frequencies below 8 MHz, good power gain above 2 MHz, and a good take-off angle for long distance communications above 10 MHz. Because of this, 40 feet is considered an ideal height. The antenna should be oriented so the long sides face distant stations.

The antenna works best when hung from two masts at least 30 and up to 40 feet above ground level. Inverted V configuration is not recommended when a low take-off angle pattern for long distance is required.

The dipole antenna described in Figure 11 utilizes an antenna coupler to operate on any frequency in the HF spectrum. It is not employed as a resonant antenna nor should it be, rather the two sets of dipole elements combine together to eliminate "bad" frequencies in which it is difficult to tune. For instance, the impedance of Elements 1R and 1L is over 5,000 ohms on certain frequencies; however, when Elements 1R/L and 2 R/L are combined, the impedance drops to about 500 ohms on those frequencies providing an easy match for the antenna coupler.

Between 2 and 8 MHz the antenna provides good overall power gain at a vertical and near vertical takeoff angles. The dipole element lengths are long enough for good performance at low frequencies and there is no significant advantage in power gain to lengthening them. Power gain on low frequencies is more defined by height above ground than element lengths.

Above 10 MHz, the elevation pattern includes enough energy at low take-off angles useful for long distance communications. As the operating frequency increases, more energy will be present at take-off angles between 30 and 10 degrees, see EZNEC plots for 2, 8, and 10 MHz.



Use of Antenna Tuners/Couplers

Antenna Tuners or Couplers are devices that are usually placed near or at the antenna, and is intended to change the impedance presented by the antenna to match that of the transmission line (see figure 11 and figure 12). In some cases, a short length of open wire type balanced line is run between the antenna coupler and the antenna, such as would be the case with a dipole directly above an antenna coupler mounted on the ground right below it. Because the coaxial transmission line between the coupler and transmitter operates at its design impedance, losses in that line are minimized, consistent across all operating frequencies, and RFI and EMI problems with equipment near the operating position are significantly reduced. Because the Antenna Coupler is usually located some distance away from the operating position, these are usually remote Automatic Antenna Tuners in weatherproof enclosures.



Figure 12 Two wire, non-resonant HF dipole antenna system with antenna coupler on ground immediately below the antenna

Utilizing a remote, automatic antenna coupler located at or near the antenna is a best practice for HF communications systems.

Broadband Dipole Antenna

Two and three wire broadband, folded dipole antennas provide a simplified way to achieve operation over a majority of the 2-30 MHz spectrum without tuning devices (see fig 13). The folded nature of the antenna increases the impedance and subsequently the bandwidth. A resistor in the parallel element attenuates the reflected wave and presents a relatively constant impedance through a transformer to the transmission line and transmitter.

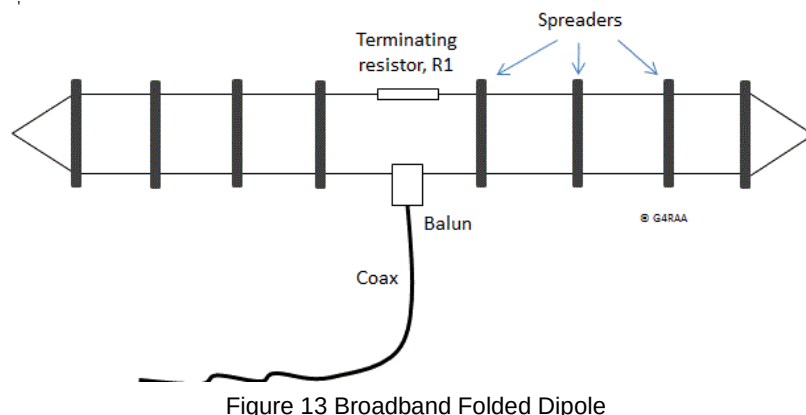


Figure 13 Broadband Folded Dipole

These antennas are approximately 50% efficient or better, provided the end-to-end length of the antenna (not total length of wire folded back on its self) is $\frac{1}{2}$ wavelength or more at the operating frequency. On operating frequencies where the antenna is shorter than $\frac{1}{2}$ wavelength, there is exponentially increasing loss in the resistor causing the antenna to be very inefficient.

is used (usually 600:50 ohms) and the antenna is at least $\frac{1}{2}$ wavelength long at the lowest planned operating frequency where good efficiency is expected (for instance an antenna 90 feet from end to end will be efficient from 5 MHz up).

Good standing wave ratio (SWR) performance is a function of the quality of the transformer, coax line, the height above ground, and the relationship to nearby objects. Antennas work best when elevated high and in the clear of other objects, and when located close to other objects are subject to unpredictable results. These types of antennas are typically used for automatic link establishment eliminating the need for an antenna coupler.

Take Off Angle

Take-off angle is the specific angle in degrees that is required to establish HF radio communications using sky wave propagation between two locations (see fig 14). The take-off angle drives the selection of the antenna system utilized for any particular radio circuit.

Take off angles less than 10 degrees are typically only attainable with horizontal antennas mounted very high above ground, or vertical antennas over sea-water. When a long-distance radio circuit requires a takeoff angle lower than what can be achieved with the antenna system, then a multiple hop path at a higher take off angle is required. These paths are subject to greater losses due to additional reflection(s) from the earth, and less availability throughout the day due to the availability of the higher critical angle at the ionosphere.

MARS stations are expected to operate on HF circuits up to 1000 miles, requiring antenna systems with sufficient energy at take-off angles as low as 15 degrees on higher frequencies.

Take-off Angle (Degrees)	Distance			
	F2 Region Daytime		F2 Region Nighttime	
	kilometers	miles	kilometers	miles
0	3220	2000	4508	2800
5	2415	1500	3703	2300
10	1932	1200	2898	1800
15	1450	900	2254	1400
20	1127	700	1771	1100
25	966	600	1610	1000
30	725	450	1328	825
35	644	400	1127	700
40	564	350	966	600
45	443	275	805	500
50	403	250	685	425
60	258	160	443	275
70	153	95	290	180
80	80	50	145	90
90	0	0	0	0

Figure 14 Take-Off Angle required for HF communications

Ideal Height for NVIS antenna

At heights greater than 30 feet, the dipole is suitable for both long distances and short-range communications. The picture in Figure 15 illustrates a real-world employment of an improvised dipole antenna at a much lower height.

Operating a dipole antenna at low heights reduces the efficiency of the communications system by introducing losses associated to proximity to the ground.



Figure 15 Improvised, non-resonant NVIS dipole antenna at low height

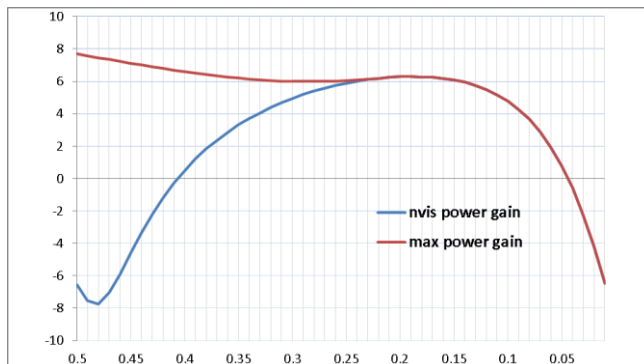


Figure 16 Power Gain as a function of Antenna Height above ground level

When employed in this manner, particularly with dismounted radio sets operating at relatively low power (20 watts), communications over relatively short NVIS paths (less than 50 miles) at night are possible in spite of the significant losses associated with using lower frequencies close to the ground. Reliable longer distance communications is not usually possible in this condition.

Figure 16 illustrates power gain in dB along the vertical axis, and antenna height above ground in wavelength along the horizontal axis. It demonstrates that radio frequency energy radiated from a horizontal antenna is absorbed by the ground when the antenna is closer than 1/10 a wavelength to the earth's surface. This absorption increases significantly as the antenna is

closer to the ground. Radio frequency radiation in the vertical or near vertical take-off angle also decreases as the antenna is elevated above 1/3 wavelength above ground, but to a much lesser extent.

Figure 17 provides the height in feet for 1/3 and 1/10 wavelength for each frequency band where NVIS operation occurs. Lower frequency bands are required for night time operation, a height of approximately 40 feet is required for efficient operation. Likewise, approximately 40 feet is the highest height above ground suitable for the higher frequencies used during the day. This chart also demonstrates that lower heights may be utilized at the expense of efficiency on the lower frequencies used at night, with very little adverse impact on daytime operations.

Height above ground is the most significant factor affecting antenna power gain and take-off angle. Although antennas at low height above ground are subject to significant power losses, they do work for short distance, single hop NVIS when there is good frequency selection and surplus power in the fade margin. Long distance communications should not be expected due to poor radiation at low take off angles, and insufficient power to travel great distances.

Frequency Band (kHz)		1/3 Wavelength (Feet)		1/10 Wavelength (Feet)	
from	to	Maximum antenna height above ground for NVIS		Minimum antenna height above ground for NVIS	
2,194.00	2,495.00	149.5	131.5	44.9	39.4
2,505.00	2,850.00	131.0	115.1	39.3	34.5
3,155.00	3,400.00	104.0	96.5	31.2	28.9
4,000.00	4,063.00	82.0	80.7	24.6	24.2
4,438.00	4,650.00	73.9	70.6	22.2	21.2
4,750.00	4,995.00	69.1	65.7	20.7	19.7
5,005.00	5,450.00	65.6	60.2	19.7	18.1
5,730.00	5,950.00	57.3	55.1	17.2	16.5
6,765.00	7,000.00	48.5	46.9	14.5	14.1
7,300.00	8,195.00	44.9	40.0	13.5	12.0

Figure 17 NVIS antenna height above ground relative to operating frequency

Broadband Inverted V Antenna

Commonly known as the RF-1944 antenna, after the Harris part number, this antenna has a long history as a easily deployed and useful antenna and has been offered by several manufacturers over the last 50 years or more. The antenna operates independent of any frequency band, throughout the range 2-30 MHz (see fig 18).

The antenna resembles a common, inverted V dipole antenna with two notable exceptions. The feed point utilizes a 450:50 Ohm (9:1) or 600:50 Ohm (12:1) transformer, and each end is grounded through a resistor of approximately 600 Ohms. The antenna is easily employed in the field, is forgiving, and presents a SWR usually less than 2.5:1 throughout the entire HF spectrum which is sufficient to use with military/commercial grade transceivers without an antenna tuner (see fig 19).

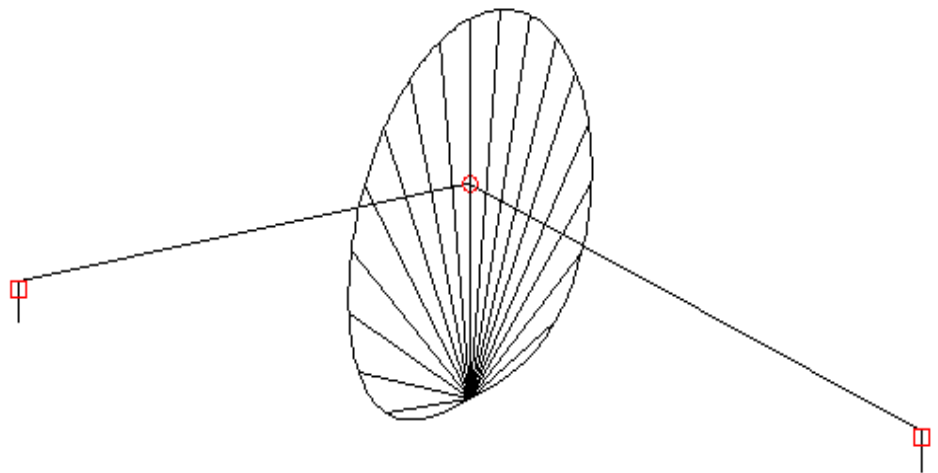


Figure 18 Broadband Inverted V (RF1944) expected radiation pattern



Figure 19 RF-1944 tactical field deployment

The antenna should be at least one-half wavelength long on the lowest operating frequency that efficient operation is intended. Like most terminated designs, operation on lower frequencies is possible at the expense of reduced antenna gain. Military versions of this antenna usually come with two 75-foot legs, for 150-foot overall length. This seems to be a good compromise between effective operation and portability.

The strength of this antennas is not in its RF performance. A low angle of radiation on higher frequencies is hard to achieve because of the inverted V configuration, and on lower frequencies, gain is between 6 to 10 dB below a resonant, half wave dipole supported between two masts of equal height to the inverted V's center mast.

The primary advantage of this antenna is that it provides a useful antenna for short to moderate range broadband HF operations, in a man portable package that is easy to employ, that is not sensitive to frequency changes. It is useful for scanning systems such as ALE where good frequency selection makes up for loss in antenna efficiency. Finally, an antenna kit that is made up of, some wire, a 9:1 or 12:1 transformer and two 600 Ohm resistors can be assembled into other configurations that provide the low take off angle, directivity and gain required for long distance radio shots.

Long Wire and V-Beams

The V Beam is a well-known broadband and directional antenna system that functions independently of any particular frequency band. It consists of the same primary elements as the Broadband Inverted V antenna discussed above, the 450:50 or 600:50 Ohm transformer, a pair of 600 Ohm resistors, with one exception: it requires much more wire (see figs 20 and 21).

One common misconception is that a Broadband Inverted V can be simply assembled with its two wires arranged into a horizontal “sloping” V to achieve gain and directivity. The V-Beam antenna requires several wavelengths of wire in each leg to achieve directivity and gain. A minimum of 3 wavelengths on the lowest frequency, and ideally 10 wavelengths means that 300-foot legs are a practical minimum, with 500 foot or more in each leg to really achieve good performance on frequencies below 10 MHz (see figure 20).



Figure 20 V Beam employment

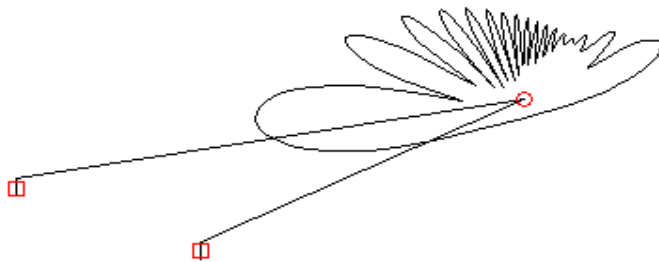
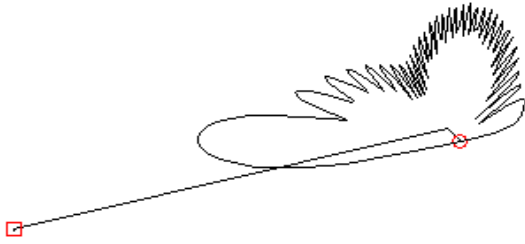


Figure 21 V-Beam expected radiation pattern

A variation and simplification of the V Beam is a Long Wire (see figure 22). It is again important to emphasize that this long wire contains a wire that is several wavelengths long. This is an antenna that is a simple wire 800 to 1600 feet long, that provides directivity, gain and a low take off angle similar to a large rotatable log periodic dipole array (LPDA) mounted on a 100-foot tower. This antenna trades the expense and complexity of the tower mounted beam for a simple wire stretched out over a very long distance. There is no free lunch with antennas, just trading requirements for something that you can support.



The long wire may be terminated for broadband operation and when terminated with a 600 Ohm resistor and fed with the 450:50 Ohm transformer (see fig 23), it becomes very directional in only one direction. When the wire is long enough, there is no penalty in gain for the terminating resistor. The longer the wire is, the lower the angle of radiation will be. Think of this as a “Beverage” antenna with a resistor and transformer rated for transmitting.

Figure 22 Very Long Wire expected radiation pattern



Figure 23 Matching Transformer (RF1944) in long wire configuration

ACRONYMS

AC - Alternating Current	HF – High Frequency
ADP – Army Doctrine Publication	HQDA – Headquarters Department of Army
ADRP – Army Doctrine Reference Publication	LPDA – Log Periodic Dipole Array
ALE – Automatic Link Establishment	MARS – Military Auxiliary Radio System
AM – Army MARS	MGEG – MARS Governance Executive Group
AR – Army Regulation	MHz - megahertz
ARCYBER – Army Cyber	MilStd – Military Standard
ATO – Authority to Operate	MSC – Military Standard Communications
CONUS – Continental United States	NETCOM – Network Enterprise Technology Command
CUI – Controlled Unclassified Information	NORTHCOM – Northern Command
dB - decibel	NSN – National Support Net
DC – Direct Current	NTIA – National Telecommunication Information Agency
DHS – Department of Homeland Security	OPSEC – Operational Security
DoD – Department of Defense	PII – Personally Identifiable Information
DoDI – Department of Defense Instruction	PSTN – Public Switched Telephone Network
DSCA – Defense Support to Civil Authorities	RF – Radio Frequency
EMI – Electromagnetic Interference	RFI – Radio Frequency Interference
EZNEC – Easy Numerical Electromagnetic Code	RJMOG – Regional Joint MARS Operations Group
FCC – Federal Communications Commission	SHARES – SHARed RESources
FEMA – Federal emergency Management Agency	STRATCOM – Strategic Command
FOUO – For Official Use Only	SWR – Standing Wave Ratio
GHFEN – Global High Frequency Enterprise Network	USC – United States Code
GMRS – General Mobile Radio System	XO – Executive Officer
HADR – Humanitarian Assistance Disaster Relief	

